Chapter 1 Preliminaries: Some of Program

1.1 Principle of Equivalence

Chapter 2 Set-Theoretic Discussions

2.1 Small, Large, Locally Small

2.1.1 Small and Large Category

Definition 2.1.1.1 Let C be a category:

- If $Ob(C)$, $Mor(C) \in Sets$, then it is called **small category**;
- Elsewise, is large category;

♣ **Example 2.1.1.2**
Large categories:

- Set: Russuel's Paradox;
- Grp;
- Top;
- $Ring$;

2.1.2 Locally Small

Definition 2.1.2.1 Especially, a category $\mathcal{C}$ is called **locally small**, if

- $Hom_C(X, Y) \in Set$;

2.1.3 Essentially Small

Definition 2.1.3.1 We use the terminology 'skeleton' and 'equivalence of categories';
TFAE:

- There exists a small category $\mathcal{D}$ such that

$$\mathcal{C} \simeq \mathcal{D}$$

- $\mathcal{C}$ has a skeleton of small category;

Then, such category is called **essentially small**.

2.2 Universe

2.2.1 Groethendieck's Universe

2.2.2 Universe Hierarchy

2.2.3 Universe Enlargement

2.3 Small Structure

2.3.1 Large and Small Limits

Chapter 3 Fundamentals

3.1 Definitions

3.1.1 Categories

Definition 3.1.1.1 A category C consists of the following elements:

- Objects $Ob(C)$;
- Morphisms $Mor(C)$,equipped with the mapping

$$s : Mor(C) \to Ob(C) \qquad t : Mor(C) \to Ob(C)$$

with:

- Composition:

$$\circ : Mor(B, C) \times Mor(A, B) \to Mor(A, C)$$

- Identity:

$$id_A \in Mor(A, A)$$

with

$$f \circ id_A = f, f \in Mor(A, B)$$
$$id_A \circ f = f, f \in Mor(B, A)$$

- Associativity

$$(f \circ g) \circ h = f \circ (g \circ h)$$

3.1.2 Functors

Definition 3.1.2.1 Let C, D be two categories.
Then, the functor is given by: $F : C \to D$ such that: For $s : C_1 \to C_2$,

$$F_t : F(C_1) \to F(C_2)$$

Notice

♣ **Example 3.1.2.2** Although sometimes could be

$$F, G : C \to D \qquad F(C) = G(C)$$

But it is still possible that

$$F \neq G$$

For example: $C = [1]$, we could obtain G from F by exchange the direction of arrow.

Full,Faithful

Definition 3.1.2.3 A functor $F : X \to Y$

$$F_{X,Y} : Hom_C(X,Y) \to Hom_D(FX, FY)$$

is called

- **full** if surjective;
- **faithful** if injective;
- **faithfully full** if bijective;

♠ **Proposition 3.1.2.4** (Criteria for Faithful) The followings are equivalent:

- F is faithful;
- $\Delta_F : C \to C \times_D C$ is full;

∘ **Proof** ... □

♠ **Proposition 3.1.2.5** (Criteria for Full) The followings are equivalent:

- F is full;
- ∇_F : is faithful;

∘ **Proof** ... □

Pre- and Postcomposition

Definition 3.1.2.6 Let $f : x \to y$ be in

- $f_*(c) : Hom(c,x) \to Hom(c,y), g \mapsto f \circ g$;
- $f^*(c) : Hom(y,c) \to Hom(x,c), g \mapsto g \circ f$;

♠ **Proposition 3.1.2.7** TFAE:

- $f : x \to y$ is an isomorphism;
- $f_*(c) : Hom(c,x) \to Hom(c,y)$ is bijection for all c;
- $f^*(c) : Hom(y,c) \to Hom(x,c)$ is bijection for all c;

∘ **Proof** ⇒
⇒
⇒ □

3.1.3 Natural Transformation

Definition 3.1.3.1 Consider the natural transformation $F : \mathcal{C} \to \mathcal{D}$ is **natural isomorphism** if there exists functor $G : \mathcal{D} \to \mathcal{C}$ such that

$$GF = id_{\mathcal{C}} \qquad FG = id_{\mathcal{D}}$$

◇ **Lemma 3.1.3.2** Consider a natural transformation $\eta : F \to G$, where $F, G : \mathcal{C} \to \mathcal{D}$.
Then, the followings are equivalent:

- η is a natural isomorphism
- $\eta(x) : F(x) \to G(x)$ are isomorphisms, $x \in X$.

∘ **Proof** Definition. □

3.1.4 Adjunction

Definition 3.1.4.1 Let $\mathcal{C}, \mathcal{D}$ be categories. pair of functors

$$F : \mathcal{C} \to \mathcal{D} \qquad G : \mathcal{D} \to \mathcal{C}$$

are called **adjunction**, if there exists a natural bijection

$$\Phi_{x,y} : Hom_{\mathcal{C}}(Fx, y) \to Hom_{\mathcal{D}}(x, Gy)$$

i.e. a natural isomorphism of functors

$$H_1 : \mathcal{C}^{op} \times \mathcal{D} \to Set \qquad (x, y) \mapsto Hom(Fx, y)$$
$$H_2 : \mathcal{C}^{op} \times \mathcal{D} \to Set \qquad (x, y) \mapsto Hom(x, Gy)$$

$$\mathcal{C}^{op} \times \mathcal{D} \quad \overset{\Downarrow}{\rightrightarrows} \quad Set$$

We will take the following notion

$$F \dashv G$$

To check the naturality, we should check the first and second variable:

- $u : X' \to X, f : FX \to Y$,

$$\Phi_{X',Y}(f \circ F(u)) = \Phi_{X,Y}(f) \circ u$$

- $v : Y \to Y', f : FX \to Y$,

$$\Phi_{X,Y'}(v \circ f) = G(v) \circ \Phi_{X,Y}(F)$$

This implies, that

$$\Phi_{X',Y'}(v \circ f \circ F(u)) = G(v) \circ \Phi_{X,Y}(f) \circ u$$

♠ **Proposition 3.1.4.2**

- Composition of left-adjoint functors is left-adjoint;
- Composition of right-adjoint functors is right-adjoint;

∘ **Proof**

-
-

□

Remark Naturality is necessary. There exists some example, that

$$\Phi_{X,Y} : Hom_C(Fx, y) \cong Hom_D(x, Gy)$$

Let G, H be two groups

$$\beta : BG \to BH, g \mapsto id \qquad \alpha : BH \to BG \qquad h \mapsto id$$

Then,

$$Hom_C(\alpha *, *) \cong G; Hom_D(*, \beta *) \cong H$$

If there exists bijection $G \to H$, or furthermore $G = H$,

$$\Phi : Hom_C(\alpha *, *) \cong Hom_D(*, \beta *)$$

But this might be not natural: If natural, then

$$\Phi(uk\alpha(v)) = \beta(u)\Phi(k)v \qquad u \in G, v \in H, k \in G$$

when α, β are trivial:

$$\Phi(uk) = \Phi(k)v$$

when $v = e_H$,

$$\Phi(uk) = \Phi(k)$$

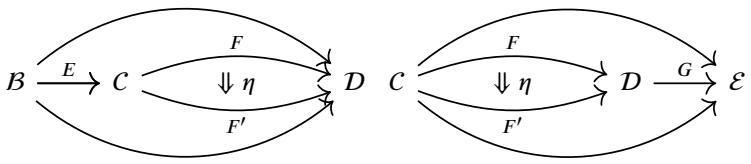
when $G = H, \Phi = id_G$, then

$$\Phi(\bar{0} \cdot \bar{1}) \neq \Phi(\bar{1})$$

Whiskering

Definition 3.1.4.3 Let $F, F' : \mathcal{C} \to \mathcal{D}, E : \mathcal{B} \to \mathcal{C}, G : \mathcal{D} \to \mathcal{E}$.
Now, given natural transformation $\eta : F \to F'$, we obtain

$$\eta * E : FE \to F'E \qquad \eta_{E(x)} : FE(x) \to F'E(x), x \in \mathcal{B}$$
$$G * \eta : \qquad : GF \to GF'$$



Unit;Counit

Definition 3.1.4.4 Let C, D be categories, consider a pair of functors

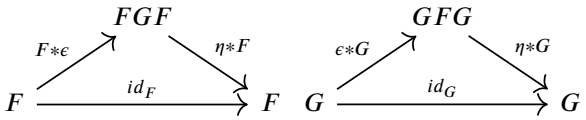
$$F : C \to D \qquad G : D \to C$$

then, the natural transformation

$$\epsilon : id_C \to GF \qquad \eta : FG \to id_D$$

are called **unit** and **counit** of F, G, if together they satsifying

$$(\eta * F) \circ (F * \epsilon) = id_F$$
$$(\eta * G) \circ (G * \epsilon) = id_G$$



Remark Here, $\circ$ is the composition of natural transformations, $*$ is **whiskering**.

♠ **Proposition 3.1.4.5** The followings are equivalent:

- There exists an adjunction pair (F, G);
- ϵ, η exists.

∘ **Proof**

- $\Rightarrow$
- $\Leftarrow$

□

Limits, Colimits

▶ ***Theorem*** In general, we could say that:

- The left adjoint functor preserves colimits;
- The right adjoint functor preserves limits;

But precisely to say, we could describe it by indexing category: Suppose there exists

$$L : C \underset{\leftarrow}{\rightarrow} D : R \qquad L \dashv R$$

Then, for any category J, and diagram $F : J \to C, G : J \to D$,

- If $colim_J F \in C$ exists, then $colim_J (L \circ F)$ exists, and

$$L(colim_J F) \cong colim_J (L \circ F)$$

- If $\lim_J G$ exists, then $\lim_J (R \circ G)$ exists, and

$$R(\lim_J F) = \lim_J (R \circ F)$$

∘ **Proof**

- Dual of limit.
- Existence: Take $l = \lim_J F$, then by limit of $Hom(A, -)$

$$Hom_{C'}(c', Rl) \cong Hom_C(Lc', l) \cong \lim Hom_C(Lc', F(d)) \cong \lim Hom_{C'}(c', RF(d))$$

Then, we obtain by universal property, $Rl = \lim_J (R \circ F)$.

□

Adjoint Functor Theorem

▶ ***Theorem*** Sufficient Conditions for a Limit-preserving functor $R : \mathcal{C} \to \mathcal{D}$ to be a **right adjoint** include:

- (**GAFT**, General Adjoint Functor Theorem)$\mathcal{C}$ is **complete** and **locally small**, and R satisfies set condition;
- (**SAFT**, Special Adjoint Functor Theorem)$\mathcal{C}$ is **compelete**, **locally small**, **well-powered**, and as a **small co-generating set**, and D is **locally small**. This

---

∘ **Proof** ... □

▶ ***Theorem*** Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between **locally presentable categories**, then

- F has a right adjoint⇔ It reverse all small colimits;
-

∘ **Proof** ... □

3.1.5 Functor Category, and Power, Copower

Functor Category

Definition 3.1.5.1 Let $\mathcal{C}, \mathcal{D}$ be two categories. Take the notion of

$$Fun(\mathcal{C}, \mathcal{D}) = [\mathcal{C}, \mathcal{D}]$$

whose morphisms are natural morphisms $Nat(F, G) \subseteq Mor(Fun(C, \mathcal{D}))$.

Power and Copower

(This power/copower is not the power in closed monoidal category, the 'internal hom')

Definition 3.1.5.2 Let C be a locally small category

- Define: The power/cotensor by a set

$$A^S \in C \qquad C(-, A^S) \cong Set(S, C(-, A))$$

- The copower/tensor by a set is

$$C(S \odot A, X) \cong Set(S, C(A, -))$$

♠ **Proposition 3.1.5.3** (Criteria for Existence, Structure of Power/Copwer) If C is Set-enriched, then

- Power exists⇔ Product exists;
- Copower exists⇔ Coproduct exists;

And

...

∘ **Proof** ... □

♠ **Proposition 3.1.5.4** (Adjointness for Hom) Suppose those power/copwer exist. We obtain, that

$$B^{(-)} \dashv Hom(-, B) \qquad - \odot A \vdash Hom(A, -)$$

∘ **Proof** ... □

3.1.6 Equivalence of Categories

'Principle of Equivalence'

`https://ncatlab.org/nlab/show/principle+of+equivalence`

Definition 3.1.6.1

- One object $A \in C$ is equivalent to another object $B \in C$, every context with A can be replaced by B.
- In higher category theory, an **equivalence** often means a morphism which is invertible in a maximally weak sence.

  Two objects are **equivalent** if there is an equivalence between them.

Isomorphism of Categories

Definition 3.1.6.2 $F : C \overset{\rightarrow}{\leftarrow} D : G$ is called **isomorphism of categories** if

$$FG = id_D, GF = id_C$$

Strong Equivalence

Definition 3.1.6.3 $F : \mathcal{C} \leftrightarrow \mathcal{D} : G$ is a **strong equivalence**, if

$$\epsilon : id_c \overset{\cong}{\rightarrow} GF \qquad \eta : FG \overset{\cong}{\rightarrow} id_D$$

Adjoint Equivalence

Definition 3.1.6.4 Let $F : C \leftrightarrow D$ be an adjunction.

- The adjunction is called an **adjoint equivalence**, if the unit and counit are natural isomorphisms. i.e.

$$\epsilon : id_c \xrightarrow{\cong} GF \qquad \eta : FG \xrightarrow{\cong} id_D$$

  In this case, F and G are called **equivalences of categories**.

Skeleton

Definition 3.1.6.5

- A strict category is called **skeletal** if for $a, b \in C<$

$$a \cong b \Rightarrow a = b$$

-
-

▶ ***Theorem*** (Existence of Skeletons of Categories)

∘ **Proof** ... □

3.1.7 Product Category

Definition 3.1.7.1 For C and D two categories. The **product category** $C \times D$ is the category:

- Objects are the **pairs**

$$(c, d) \qquad c \in C, d \in D$$

  written as $Obj(C \times D) = Ob(C) \times Ob(D)$;

- Morphisms

$$(f : c \to c', g : d \to d') \qquad Mor(C \times D) = Mor(C) \times Mor(D)$$

- Composition of Morphisms, defined componentwie by composition in C and D.

◇ **Lemma 3.1.7.2** The following types are equivalent:

- Functors $A \times B \to C$;
- Functors $A \to C^B$.

∘ **Proof** Omitted. □

Thus,

♠ **Proposition 3.1.7.3** We obtain an isomorphism of categories

$$Fun(I, C) \cong C^I$$

where I could be the discrete category of cardinal ω. This is the Cartesian product.

∘ **Proof** Given by $F(1) = X \in C, F(2) = Y \in C$. □

3.1.8 Generating Objects

Definition 3.1.8.1 Let C be a category. An object G is called a **generator** if it satisfies the following property: For $f, g : X \to Y; f \neq g$, then there exists a morpism from generator G,

$$h : G \to X$$

such that $f \circ h \neq g \circ h$.

This implies, that the performance of h decides the performance of X.

3.1.9 Join

Definition 3.1.9.1 Let A, B be categories, then define:

- $Ob(A \star B) = Ob(A) \sqcup Ob(B)$
- $Hom_{A\star B}(x, y) = \begin{cases} Hom_C(x, y) & x, y \in A \text{ or } x, y \in B \\ \{*\} & x \in A, y \in B \\ \emptyset \end{cases}$

3.2 Limits, Colimits

3.2.1 Cone, Cocone

Cone Structure in Category

Definition 3.2.1.1

- The **cone** of C is the cocomma object of identity on C, and the unique map to the terminal category $! : C \to *$

$$\begin{array}{ccc} C & \xrightarrow{id_C} & C \\ \downarrow{\scriptstyle !} & \Rightarrow & \downarrow \\ * & \longrightarrow & Cone(C) \end{array}$$

- Again, this may be computed as a pushout

$$\begin{array}{ccc} C & \xrightarrow{d_1} & C \times 2 \\ \downarrow & & \downarrow \\ * & \longrightarrow & Cone(C) \end{array}$$

Cone over Index Category

Definition 3.2.1.2 A **cone** in a category is given by a category J, together with a functor

$$cone(J) \to C$$

By the universal property,

3.2.2 Limits, Colimits

Set-Valued Functor Limit

Definition 3.2.2.1 Suppose D be small category.

- Given a functor $F : D^{op} \to Set$, then

$$\lim F := Hom_{[D^{op},Set]}(pt, F) \in Set$$
$$pt : D^{op} \to Set \qquad d \mapsto \{*\}$$

There is no such construction of equalizer. Could specified by

$$\text{colim}_D \dashv \Delta \dashv \lim_D \qquad \Delta : C \to [D, C]$$

♠ **Proposition 3.2.2.2** (Limit, Colimit as Equalizer, Coequalizer) We obtain, that for $F : I \to C$

$$\lim_I F = Eq(\prod_i F_i \overset{s,t}{\to} \prod_{u:i\to j} F_j)$$
$$\text{colim}_I F = Coeq(\sqcup F_i \overset{s,t}{\to} \sqcup_{u:i\to j} F_j)$$

∘ **Proof** Omitted. We will prove this for equivalence: Compeleteness and Cocompeleteness. □

Set-Theoretic Discussion

♠ **Proposition 3.2.2.3** There is bijection

$$Hom_{set}(S, limF) \cong \lim Hom_{Set}(S, F(-))$$

∘ **Proof** ... □

Limit of a Functor, with Values in Arbitrary Category

Definition 3.2.2.4 If C is locally small category:

- By Yoneda embedding $y : C \to [C^{op}, Set]$, define for $F : D^{op} \to C$, the presheaf

$$(\hat{lim}F) \in [C^{op}, Set] \qquad (\hat{lim}F)(c) = Nat(pt, Hom(c, F(-))) = Nat(pt, y(F(-))(c))$$

- By co-Yoneda embedding $y^{\vee} : C^{op} \to [C, Set], c \mapsto Hom(c, -)$, define for $F : D^{op} \to C$,

$$\hat{colim} F \in [C, Set] \qquad \hat{colim} F(c) := Nat(pt, Hom_C(F(-), c)) = Nat(pt, y^{\vee}(F(-))(c))$$

♠ **Proposition 3.2.2.5** (Existence)

- The presheaf $\hat{lim}$ exists, if it is representable by an object $\lim F \in C$,

$$Hom(c, limF) \cong Hom(pt, Hom(c, F(-))) \Rightarrow Hom(-, \lim F) \cong (\hat{lim} F)$$

- The presheaf $\hat{\text{colim}}$ exists, if it is representable by an object $\text{colim}\, F \in C$

$$Hom(\text{colim}\, F, c) \cong Hom(pt, Hom(F(-), c)) \Rightarrow Hom(\text{colim}\, F, -) \cong (\hat{\text{colim}} F)$$

∘ **Proof** ... □

♠ **Proposition 3.2.2.6** (Right and Left Adjoint Preserving) Let D be a small category

- Let $R : C \to C'$ be a functor which is right adjoint, and C admits limits of shape D. Then, R commutes with D-shaped limits in C, in that
  For $F : D^{op} \to C$, we have

$$R(\lim F) \cong \lim(R \circ F)$$

-

∘ **Proof** Proved. □

♠ **Proposition 3.2.2.7** (Hom-Functor Preserving) For C a locally small category, functor $F : D^{op} \to C$,

- $Hom_C(c, F(-)) : D^{op} \to Set$,

$$Hom_C(c, \lim F) \cong \lim Hom_C(c, F(-))$$

- $Hom_C(F(-), c) : D \to Set$,

$$Hom_C(\text{colim}\, F, c) \cong \text{colim}\, Hom_C(F(-), c)$$

∘ **Proof**

- Define: $L := \lim F$, then $u : c \to L$,

$$\lambda_d \circ u : c \to F(d)$$

  Then, there exists a family $(\lambda_d \circ u)$, then

$$\Phi : C(c, L) \to \lim C(c, F(d)) \qquad u \mapsto (\lambda_d u)_d$$

  Conversely, for $(u_d)_d \in \lim_d C(c, F(d))$, we obtain a unique

$$u : c \to L \qquad \lambda_d \circ u = u_d$$

  Thus, we obtain

$$\Psi : \lim_d C(c, F(d)) \to C(c, L) \qquad (u_d)_d \mapsto u$$

  And $\Psi\phi, \Phi\Psi$ are inverse;

- Similarly.

□

♠ **Proposition 3.2.2.8** (Product and Coproduct Componentwised)

∘ **Proof** ... □

Universal Cone

The limit and colimit could be given by the diagram:

...

Fubini's Interchange Isomorphism

♠ **Proposition 3.2.2.9**

- (Limit) Let $F(-,-) : K \times L \to C$ be a functor such that: For every $l \in L$, there exists a limit

$$\lim_{k \in K} F(-, l) \in C$$

  Then, the followings are equivalent:

  – $F(-,-)$ admits a limit;
  – Construction of canonical functor

$$\lim_{k \in K} F(k, -) : L \to C \qquad l \mapsto \lim_{K} F(-, l)$$

    admits a limit;

  Furthermore, in this case we obtain

$$\lim_{K \times L} F(-,-) \simeq \lim_{L} \lim_{k \in K} F(k, -) \in C$$

- (Colimit) The following are equivalent:

  – $F(-,-)$ admits a colimit;
  – Construction of canonical functor

...

    admits a colimit;

∘ **Proof** ... □

★ **Corollary 3.2.2.10** If $F : K \times L \to C$ be a diagram with K, L small categories.
If F admits a limit, and the limit of $F(k,-), F(-,l)$ exists, then for all $k \in K, l \in L$,

...

∘ **Proof** ... □

And for limit and colimit, a counter example is:

♣ **Example 3.2.2.11** ...

Filtered Colimits, Finite Limits Exchanges

♠ **Proposition 3.2.2.12** ...

∘ **Proof** ... □

Terminal and Initial Objects

Definition 3.2.2.13

- The **terminal object** of a category is an object $T \in \mathcal{C}$ such that: For every object $X \in \mathcal{C}$, there exists one and only one morphism $X \to T$;
- The **initial object** of a category is an object $I \in \mathcal{C}$ such that:For every object $X \in \mathcal{C}$, there exists one and only one of morphism $I \to X$.

♠ **Proposition 3.2.2.14** The terminal and initial object are unique, up to isomorphism, if exists.

∘ **Proof** Omitted. □

♠ **Proposition 3.2.2.15** Then,

$$T \cong \lim_{\emptyset} F \qquad F : \emptyset \to \mathcal{C}$$
$$I \cong \operatorname{colim}_{\emptyset} G \qquad G : \mathcal{C} \to \emptyset$$

∘ **Proof**

•

•

□

3.2.3 Preverse of Limits and Colimits, Continuous annd Cocontinuous Functors

Definition 3.2.3.1 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor:

- F is called **continuous**, if it preserves all small limits;
- F is called **cocontinuous**, if it preserves all small colimits;

♣ **Example 3.2.3.2**

- Consider the free-forgetful adjointness:

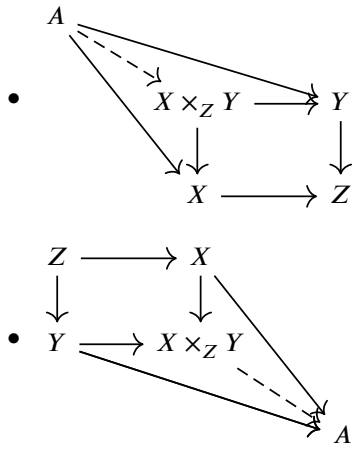
$$Free : Set \underset{\leftarrow}{\rightarrow} Grp : U \qquad Free \dashv U$$

Notice: $Free$ is cocontinuous and U is continuous;

$$Free(\{*\} \times \{*\}) \cong \mathbb{Z}$$
$$Free(\{*\}) \times Free(\{*\}) \cong \mathbb{Z} \times \mathbb{Z}$$

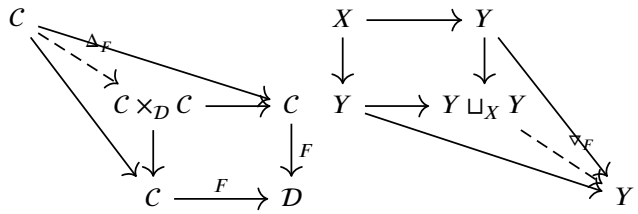
3.2.4 Pullbacks and Pushout

Definition 3.2.4.1



Relative Diagonal and Codiagonal

♠ **Proposition 3.2.4.2** With the diagonal, codiagonal, we obtain the relative diagonal and codiagonal: By



∘ **Proof** Omitted. □

3.3 Product, Coproduct

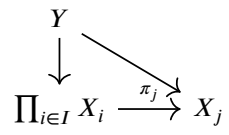
3.3.1 Definitions

Definition 3.3.1.1 Let $\{X_i\}_{i\in I}$ be a family of objects

- The product

$$\prod_{i\in I} X_i$$

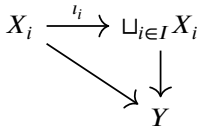
with universal property



- The coproduct

$$\sqcup_{i\in I} X_i$$

with universal property



♠ **Proposition 3.3.1.2** (Commutativity of Product, Coproducts) For index category I, J, and a family of objects $F_{i,j}$:

- $$\prod_I \prod_J F_{i,j} \cong \prod_J \prod_I F_{i,j}$$

- $$\sqcup_I \sqcup_J F_{i,j} \cong \sqcup_J \sqcup_I F_{i,j}$$

∘ **Proof**

-
-

□

Binary Product, Coproduct

Definition 3.3.1.3

- Binary Product $A \times B$;
- Binary Coproduct $A \sqcup B$;

Nullary Product and Corproduct, Existence Criteria

Definition 3.3.1.4

- Nullary product $\prod_\emptyset X = 1$;
- Nullary coproduct $\sqcup_\emptyset = 0$;

We should verify, that they satisfy the property of terminal object:

∘ **Proof**

- $I \to X$ is not needed;
- Similarly;

□

♠ **Proposition 3.3.1.5** (Universal Property with Hom)

$$Hom(X, \prod_I Y_i) \cong \prod_I Hom(X, Y_i)$$
$$Hom(\sqcup_I X_i, Y) \cong \sqcup_I Hom(X_i, Y)$$

∘ **Proof** ... □

Discussion of Associativity

There exists canonical isomorphism

$$(\pi_1 \circ \pi_1, \times \circ (\pi_2 \circ \pi_1, \pi_2)) \qquad (X \times Y) \times Z \cong X \times (Y \times Z)$$
$$(\pi_2, \pi_1) \qquad X \times Y \cong Y \times X$$
$$X \times 1 \cong X$$

Thus, $(\mathcal{C}, \times, 1)$ is a Cartesian symmetric monoidal category.

Pullback, Pushout

♠ **Proposition 3.3.1.6** ...

∘ **Proof** ... □

Finite Products

3.3.2 Cartesian Category

Definition 3.3.2.1 A **cartesian category** is a category with finite products.

We will use this concepts in the later discussion.

3.3.3 Diagonal,Codiagonal

Definition 3.3.3.1 Let $\mathcal{C}$ be a category:

- If $X \times X$ exists, then the canonical morphism

$$\Delta_X : X \to X \times X \qquad \pi_1 \Delta_X = \pi_2 \Delta_X = id_X$$

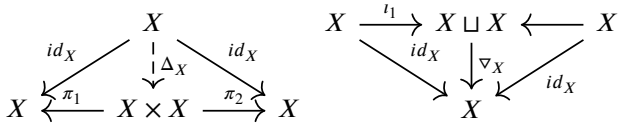
  is called the **diagonal morphism**; Denoted by $\Delta_X =< id_X, id_X >$;

- Similarly,

$$\nabla : X \sqcup X \to X \qquad \nabla_X \iota_1 = \nabla \iota_2 = id_X$$

  is called the **codiagonal morphism**; Denoted by $\nabla_X = [id_X, id_X]$;

Diagram, and universal properties



Diagonal and Codiagonal as Element Functor Categories

◇ **Lemma 3.3.3.2** Consider the index discrete category $I = \{x_1, x_2\}$, then there exists an isomorphism

$$Func(I, C) \cong C \times C$$

∘ **Proof** ... □

★ **Corollary 3.3.3.3** Then, we can view diagonal and codiagonal as the functor

$$\Delta_X : C \to Func(I, C)$$
$$\nabla_X : Func(I, C) \to C$$

∘ **Proof** ... □

Adjunciton

♠ **Proposition 3.3.3.4** There exists adjunction: For discrete category I,

$$\sqcup_I \dashv \Delta_I \dashv \prod_I$$

Especially, for $I = \{1, 2\}$

...

3.4 Functor Categories

Definition 3.4.0.1 Given categories C, D, the **functor category**, written as $[C, D]$ is the category whose:

- $Ob([C, D]) = Func(C, D)$;
- Morphsims are natural transformations.

3.4.1 I-Shaped

Definition 3.4.1.1 For fixed I, then

$$Func(I, C) = I - \text{shaped diagrams in } C$$

With:

- Objects $Ob(Func(I, C)) = \{F : I \to C\}$;
- Morphisms $Mor(Func(I, C)) = Nat(F, G)$;

For element $F : I \to C$, such I is called **index category**.

Discrete Category-Shaped

♠ **Proposition 3.4.1.2** If S is discrete category, then

$$Func(S, C) \cong C^S$$

∘ **Proof** Trivial. □

Evaluation and Diagonal

Definition 3.4.1.3 Let C be a category:

- Evaluation $ev_i : Fun(I, C) \to C, F \mapsto F(i)$;
- Diagonal/Constant $\Delta : C \mapsto Func(I, C), X \mapsto (i \mapsto X)$;

A special case is $I = \{1, 2\}$.

♠ **Proposition 3.4.1.4** Evaluation preserves limits and colimits;

∘ **Proof** ... □

★ **Corollary 3.4.1.5** ...

∘ **Proof** ... □

Limits, Colimits, Adjunction

Definition 3.4.1.6 Now suppose I be the 'index category'.

- The limit $\lim_I$ should be understand as a functor $Func(I, C) \to C$;
- The colimit colim_I should also be understand as a functor $C \to Func(I, C)$

♠ **Proposition 3.4.1.7** There exists adjunction $\text{colim}_I \dashv \Delta \dashv \lim_I$. i.e. for $F : I \to C$,

$$Hom_C(\text{colim}_I F, Y) \cong Hom_{Fun}(F, \Delta_I Y)$$
$$Hom_{Fun}(\Delta_I Y, G) \cong Hom_C(Y, \lim_I G)$$

where Δ is the diagonal mapping.

∘ **Proof** We obtain this as following

•

•

□

★ **Corollary 3.4.1.8** For $I = \{1, 2\}$, and the diagram $F : I \to \mathcal{C}$,

$$\sqcup \dashv \Delta \dashv \times$$

∘ **Proof** We just need to give $F(1), F(2)$. Thus,

$$Func(I, \mathcal{C}) \cong \mathcal{C} \times \mathcal{C}$$

□

Compeletness

Definition 3.4.1.9 Let $\mathcal{C}$ be a category:

- $\mathcal{C}$ is called **compelete**, if it contains all small limits;
- $\mathcal{C}$ is called **cocomplete**, if it contains all small colimits;

♠ **Proposition 3.4.1.10**

- TFAE:
  - Completeness;
  - $\mathcal{C}$ has all small products and equalizers;
- TFAE:
  - Cocompleteness;
  - $\mathcal{C}$ has all small products and equalizers;

∘ **Proof** We just need to prove the product: ⇒ Small Product: The limit with discrete index category; Equalizers: The equalizer could be given by the diagram

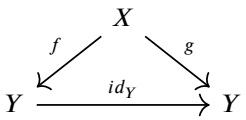
$$Eq(f, g) \dashrightarrow X \underset{g}{\overset{f}{\rightrightarrows}} Y$$

It can be expressed as

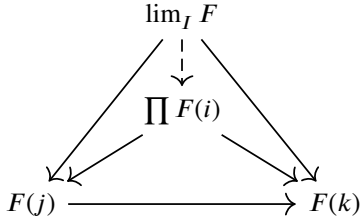
$$Eq(f, g) = \lim_I F \qquad F : I \to \mathcal{C}$$

where

$$I = (\ 0 \underset{v}{\overset{u}{\rightrightarrows}} 1\ )$$



⇐ We obtain the limit as following: Notice that



Thus, for every j, k and $F(j) \to F(k)$, we obtain

$$\lim_I F = Eq(\prod_i F(i) \overset{s,t}{\to} \prod_{u: i \to j} F(j))$$

where

$$s_u : \prod_i F(i) \to F(j) \qquad (s_i)_{i \in I} \mapsto s_{s(u)}$$
$$t_u : \prod_i F(i) \to F(j) \qquad (t_i)_{i \in I} \mapsto t_{t(u)}$$

□

Set-Theoretic Discussion

♠ **Proposition 3.4.1.11** Especially:

∘ **Proof** ... □

Exponential, Currying

▶ ***Theorem*** (Currying) There exists isomorphism of categories

$$Func(I \times J, C) \cong Func(I, Func(C, J))$$
$$(F : I \times J \to C) \mapsto (\tilde{F} : I \to Func(J, C))$$

∘ **Proof** Omitted. □

Inverse/Projective Limit and Inuductive/Direct Limit

Definition 3.4.1.12 Especially, on the partial ordered system $(I, \leq)$

- The **inductive limit**

$$\varinjlim_{i \in I} A_i = \text{colim}_{i \in I} A_i$$

- The **inverse limit**

$$\varprojlim_{i \in I} A_i = \lim_{i \in I} A_i$$

3.4.2 Restriction/Pullback of Diagrams/Precompostion

♠ **Proposition 3.4.2.1** Given: Functor $u : I \to J$, then the precomposition

$$u^* : Func(J, C) \to Func(I, C) \qquad F \mapsto F \circ u$$

∘ **Proof** Omitted. □

★ **Corollary 3.4.2.2** $I \mapsto Func(I, C)$ is contravariant for indexing category

♠ **Proposition 3.4.2.3** There exists adjunction

$$Lan_u \dashv u^* \dashv Ran_u$$

∘ **Proof** ... □

Especially: For functor $p : I \to *$, p^* is the constant diagram functor Δ. Thus

$$Lan_p = colim_I \qquad Ran_p = \lim_i$$

3.5 Comma Category, and Slice Category

3.5.1 Comma Category

Definition 3.5.1.1 Let: $f : C \to E, g : D \to E$ be functors, their **comma category** is denoted by (f/g) whose:

- $Ob(f/g)$: Triples

$$(c, d, \alpha) \qquad \alpha : f(c) \to g(d); c \in C, d \in D, \alpha \in Mor(E)$$

- $Mor(f/g)$:

$$\begin{array}{ccc} f(c_1) & \xrightarrow{\alpha_1} & g(d_1) \\ \downarrow{\scriptstyle f\beta} & & \downarrow{\scriptstyle f\gamma} \\ f(c_2) & \xrightarrow{\alpha_2} & g(d_2) \end{array} \qquad (c_1, \alpha_1, d_1) \overset{(\beta,\gamma)}{\to} (c_2, \alpha_2, d_2).$$

- Composition is composition of morphisms.

To distinguish with the cocomma category, us the notion of $F \downarrow G$.

Cocomma Category

Similarly

Definition 3.5.1.2 $F : E \to C, G : E \to D$, the **cocomma category**

- $Ob(F \downarrow_{co} G) := Ob(C) \sqcup Ob(D)$;
- $Mor(F \downarrow_{co} G) := Mor(C) \cup Mor(D) \cup \{\tau_e | e \in E\}$, where

$$\begin{array}{ccc} F(a) & \xrightarrow{F\alpha} & F(a') \\ \downarrow{\scriptstyle \tau_a} & & \downarrow{\scriptstyle \tau_{a'}} \\ G(a) & \xrightarrow{G\alpha} & G(a') \end{array} , a, a' \in E$$

♠ **Proposition 3.5.1.3** As a matter of fact,

$$(F \downarrow_{co} G) \simeq B \sqcup_A (A \times [1]) \sqcup_A C$$

∘ **Proof** ... □

Comma Object

Compeleteness,Cocompeleteness

3.5.2 Slice Category

Definition 3.5.2.1 The **slice category** $\mathcal{C}/c$ over object $c \in C$ is:

- Objects are the arrows $f \in Mor(C)$ such that

$$t(f) = c$$

- Morphisms: $g : X \to X'$ for $f : X \to c, g : X' \to c$ such that $f' \circ g = f$.

Thus,

$$C/c = \begin{array}{ccc} X & \xrightarrow{g} & X' \\ & \searrow^{f} \quad \swarrow & \\ & c & \end{array}$$

According to definition, we obtain that

$$C/c = (Id_C/F_c) \qquad F_c :* \to C \qquad * \mapsto c$$

3.5.3 Arrow Category

Definition 3.5.3.1 For C any category, its arrow category $Arr(C)$ is the category such that:

- Objects are arrows;
- Morphisms are commutative squares;
- Composition are composiiton of squares.

♠ **Proposition 3.5.3.2** Let $I = \{0 \to 1\} = 2$ be the interval category, then

$$Arrow(C) \cong Func(I, C)$$

∘ **Proof** Trivial; □

3.5.4 Twisted Category

Definition 3.5.4.1 Define: The twisted category

- $Ob(Tw(C)) = Mor(C)$;
- $Mor(Tw(C))$ are such (p, q) with following diagram commutes:

$$\begin{array}{ccc} A & \xleftarrow[p]{} & C \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} \\ B & \xrightarrow{q} & D \end{array}$$

3.6 Grothendieck Construction

3.6.1 Category of Elements

3.7 End and Coend

3.7.1 Extranatural

Definition 3.7.1.1

- A family of morphisms $\alpha_{a,b,c} : F(a,b,b) \to G(a,c,c); F : A \times B \times B^{op} \to D, G : A \times C \times C^{op} \to D$ is **ordinary-natural** in a and **extranatural** in $b$and c if the following hold:
  - For $f : a \to a'$ in A; $b \in B, c \in C$, the following square commutes:

$$\begin{array}{ccc} F(a,b,b) & \xrightarrow{F(f,id_b,id_b)} & F(a',b,b) \\ \downarrow{\scriptstyle \alpha_{a,b,c}} & & \downarrow{\scriptstyle \alpha_{a',b,c}} \\ G(a,c,c) & \xrightarrow{G(f,id_c,id_c)} & G(a',c,c) \end{array}$$

  - For all $g : b \to b'$ in B, and all $a \in A, c \in C$, the following square commutes(extranaturality in b)

$$\begin{array}{ccc} F(a,b,b') & \longrightarrow & F(a,b,b) \\ \downarrow & & \downarrow \\ F(a,b',b') & \longrightarrow & G(a,c,c) \end{array}$$

  - For all $h : c \to c'$ in C, and all $a \in A, b \in B$, the following square commutes(extranaturality in c)

$$\begin{array}{ccc} F(a,b,b) & \xrightarrow{F(\alpha_{a,b,c}} & G(a,c,c) \\ \downarrow{\scriptstyle \alpha_{a,b,c'}} & & \downarrow{\scriptstyle G(1,h,1)} \\ G(a,c',c') & \xrightarrow{G(1,1,h)} & G(a,c',c) \end{array}$$

- Especially, when $A = *$. Suppose:

$$F : C \times C^{op} \to \mathcal{D}$$

Fix a $X \in \mathcal{D}$, a family of morphisms

$$\alpha_c : F(c,c) \to X$$

is called **extranatural**, if for each $f : c \to c'$, $\alpha_c \circ F(f, 1_c) = \alpha_{c'} \circ F(1_{c'}, f)$

$$\begin{array}{ccc} F(c',c) & \xrightarrow{F(1_{c'},f)} & F(c',c') \\ \downarrow{\scriptstyle F(f,1)} & & \downarrow{\scriptstyle \alpha_{c'}} \\ F(c,c) & \xrightarrow{\alpha_c} & X \end{array}$$

Conversely,

Definition 3.7.1.2

-
- Suppose $F : C^{op} \times C \to X$, then a family of morphisms

$$\eta_c : F(c,c) \to x$$

is called **coextranatural**, if for each $f : c \to c', \eta_c \circ F(f, 1_c) = \eta_{c'} \circ F(1_{c'}, f)$

...

3.7.2 Wedge, Cowedge

Definition 3.7.2.1 Let $F : C^{op} \times C \to X$ be a functor.

- A **wedge** $e : w \to F$ is:
  - An object w;
  - Maps $e_c : w \to F(c,c)$;
  - For each c, c' in C, $f : c \to c'$, the following diagram commutes:

$$\begin{array}{ccc} w & \xrightarrow{e_{c'}} & F(c',c') \\ \downarrow{\scriptstyle e_c} & & \downarrow{\scriptstyle F(f^{op}, id_{c'})} \\ F(c,c) & \xrightarrow{F(id_c, f)} & F(c,c') \end{array}$$

- A **cowedge** $e : F \to w$ is:
  - An object w;
  - Maps $e_c : F(c,c) \to w$;
  - For each c, c' in C, $f : c \to c'$, the following diagram commutes:

$$\begin{array}{cc} F(c,c') & F(c,c) \\ \\ F(c',c') & w \end{array}$$

3.7.3 End, Coend

Definition 3.7.3.1

- Given a functor $F : C^{op} \times C \to X$, an **end** of F in X is an object e of X, equipped with a universal extranatural transformation from e to F.

  Denoted by $\int_{c:C} F(c,c)$.

  i.e.
  - Object $e = \int_{c:C} F(c,c)$;
  - Extranatural morphisms $\pi_c : e \to F(c,c)$;
- By universal property: For any wedge

$$(x, (\alpha_c)_{c \in C}) \qquad \exists u : x \to e, \alpha_c = \pi_c \circ u$$

It is actually a special kind of limit over a functor of the form of bifunctor:

♠ **Proposition 3.7.3.2**

$$\int_{c:C} F(c,c) = \lim_{Tw(C)} F \circ p \qquad p : Tw(C) \to C^{op} \times C$$

∘ **Proof** ... □

♣ **Example 3.7.3.3** We obtain, that for functors $F, G : \mathcal{C} \to \mathcal{D}$,

$$Nat(F,G) \cong \int_{c \in C} Hom_D(F(c), G(c))$$

Right and Left Adjointness

Equalizer and Coequalizer

Fubini's Theorem

Chapter 4 Groupoids

4.1 Fundamentals

Definition 4.1.0.1

- A small groupoid G is:
  - A set X, which is called the set if objects;
  - Pairs $(x, y) \in X \times X \Rightarrow Hom(x, y)$;
  - Triples $(x, y, z) \in X \times X \times X \Rightarrow \circ_{x,y,z} : Hom(y, z) \times Hom(x, y) \to Hom(x, z)$;
  - The identities

$$id_x \in Hom(x, x)$$

  - For pairs $(x, y) \in X \times X$, a function

$$(-)^{-1} : Hom(x, y) \to Hom(y, x)$$

    is the inverse-assigning function.

  such that

  - (Associativity) For x_1, x_2, x_3, x_4,

$$h \circ (g \circ f) = (h \circ g) \circ f$$

  - For $f \in Hom(x, y)$

$$id_y \circ f = f \qquad f \circ id_x = f$$

  -

$$f^{-1} \circ f = id_x \qquad f \circ f^{-1} = id_y$$

- Let $\mathcal{G}_1, \mathcal{G}_2$, then the **homotopy/natural transformation**

$$\eta : F \Rightarrow G$$

  with:

  - A function $F_0 : X_1 \to X_2$;
  - For $x, y \in X_1$, then

$$F_{x,y} : Hom(x, y) \to Hom(F_0(X), F_0(y))$$

  such that

  - (Composition)
  - (Identities)

- The **conjugation** or **homotopy** $\eta : F \Rightarrow G$:

– For each object $x \in X_1$ of $\mathcal{G}_1$, a morphism $\eta_x \in Hom(F(x), G(x))$;
– For all $x, y \in X_1$ and $f \in Hom(x, y)$, the equality

$$\eta_y \circ F(f) = G(f) \circ \eta_x$$

$$\begin{array}{ccc} F(x) & \xrightarrow{\eta_x} & G(x) \\ \downarrow{\scriptstyle F(f)} & & \downarrow{\scriptstyle G(f)} \\ F(y) & \xrightarrow{\eta_y} & G(y) \end{array}$$

And composition of homotopies

$$\eta_2 \circ \eta_1 : F \Rightarrow H$$

4.1.1 'Base Loops' of Category:Base Loop Group

Definition 4.1.1.1 Let C be a category, with object $x \in C$.
Then, define

$$\Omega_x C := End_C(x) = Hom_C(x)$$

4.1.2 Deloping Category

Definition 4.1.2.1 The **delooping** of an object $A \in C$, it is exists, a uniquely pointed object $BA \in C$ such that: A is the loop space object of BA:

$$A \simeq \Omega(BA)$$

♠ **Proposition 4.1.2.2** Suppose G be a group, then the space BG exists.

∘ **Proof** Given by:

- $Ob(BA) = \{*\}$;
- $End_C(*) = G$

And $G \simeq \Omega(BG)$: □

And:

♠ **Proposition 4.1.2.3** Let G be a group, which is small, then

- Category $Fun(BG, Set)$;
- The category of sets with:
  – A G-action;

$$\begin{gathered} G \times BG \to BG \\ (g, *) \mapsto * \\ (g, x) \mapsto Bg \circ x \end{gathered}$$

And satisfies

$$\rho(gh, x) = \rho(g, \rho(h, x)) \qquad \rho(e, x) = x$$

– Equivariant Maps:For the map $f : X \to Y$,

$$\rho_X : G \times X \to X \qquad \rho_Y : G \times Y \to Y$$
$$f(g \cdot x) = g \cdot f(x)$$

This implies, that

$$Fun(BG, Set) \cong G - Set$$

∘ **Proof** $\Rightarrow$ The action $\rho : G \times BG \to BG$ is specified by

$$(g, f) \mapsto Bg \circ f$$

where $B : G \to Mor(BG)$. And it is easy to verify that

$$\rho(gh, x) = \rho(g, \rho(h, x)) \qquad \rho(e, x) = x$$

And the equivariance map is specified by the natural transformations: Suppose $F : h \Rightarrow g$, then

$$F(\rho(g, x)) = F(Bg \circ x) = F(Bg) \circ F(x) =$$

$\Leftarrow$ Consider the G-set

$$(X, \rho)$$

Then, the mapping $F_X \in Fun(BG, Set)$ is specified by

$$BG \to Set \qquad * \mapsto X$$
$$Mor(BG) \to Mor(Set) \qquad g \mapsto (x \mapsto g \cdot x)$$

And satisfies

$$F_X(gh) = F_X(g) \circ F_X(h)$$

And the morphisms $f : X \to Y$ induces the natural transformation

$$\eta_f : F_X \Rightarrow F_Y \qquad \eta_f(F_X(x)) := F_Y(f(y))$$

which satisfies $\eta_f(g \circ h) = \eta_f(g) \circ \eta_f(h)$. □

★ **Corollary 4.1.2.4** Furthermore,

$$Fun(BG^{op}, Set) \cong Set - G$$

∘ **Proof** Similarly □

Furthermore, we obtain that

♠ **Proposition 4.1.2.5** Let G be a group. Then,

$$Nat(Id_{BG}, Id_{BG}) \cong Z(G)$$

∘ **Proof** Now, consider the diagram

$$BG \;\underset{\overset{id_{BG}}{\Downarrow \eta}}{\overset{id_{BG}}{\rightrightarrows}}\; BG$$

which maps $*$ to $*$, and for $g \in End(*)$,

$$\eta_* \circ id_{BG}(g) = id_{BG}(g) \circ \eta_* \Rightarrow \eta_* \circ g = g \circ \eta_*$$

$$\begin{array}{ccccc} * & \xrightarrow{id_{BG}} & * & \xrightarrow{\eta_*} & * \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle id_{BG}(g)} & & \downarrow{\scriptstyle id_{BG}(g)} \\ * & \xrightarrow{id_{BG}} & * & \xrightarrow{\eta_*} & * \end{array}$$

This implies, that η_* is the commutator of G.

$$Nat(Id_{BG}, Id_{BG} \cong Z(G)$$

□

4.2 Nerve Constructions

Definition 4.2.0.1 Suppose C, then, the **nerve**

$$N(C) : \Delta^{op} \to Set$$

where

$$Ob(\Delta) = \{[n] | n \in \mathbb{N}\} \qquad Hom_{[n]}(i,j) = \begin{cases} \{*\} & i \leq j \\ \emptyset & i > j \end{cases}$$

♠ **Proposition 4.2.0.2** (Strict Segal Condition)

$$N(C)_n \cong \underbrace{N(C)_1 \times_{N(C)_0} \times_{N(C_0)} N(C)_1}$$

∘ **Proof** ⇒
⇐ □

Definition 4.2.0.3 A simplical set satisfies **Segal conditions**, if the Segal maps exhibit equivalences

$$X_n \overset{\simeq}{\to} X_1 \times_{X_0} \times \cdots \times_{X_0} X_1$$

Furthermore, we have the following theorem, due to Grothendieck

▶ ***Theorem*** (Grothendieck) TFAE:

- A simplicial set is the nerve of a category;
- It satisfies Segal conditions;

∘ **Proof** ... □

4.2.1 Internal Categories

Definition 4.2.1.1 Let $\mathcal{E}$ be a category with (small) limit. Then, the **internal category** is

$$X : \Delta^{op} \to \mathcal{E}$$

such that

$$X_n \cong \underbrace{X_1 \times_{X_0} \cdots \times_{X_0} X_1}_{n}$$

We will generalize this in ∞-category.

4.2.2 Groupoid

▶ ***Theorem*** TFAE:

- C is a groupoid;
- $N(C)$ is a Kan complex;

∘ **Proof** $\Rightarrow$
$\Leftarrow$ □

Chapter 5 Presheaves, Yoneda Lemma

5.1 Presheaves

Definition 5.1.0.1 Let C be a category:

- A **presheaf** is a functor

$$\mathscr{F} : C^{op} \to Set$$

- The category of presheaf is denoted by $Psh(C)$;
- A **copresheaf/covariant** is a presheaf on the opposite category C^{op}

$$:C \to Set$$

5.1.1 Representable Presheaf

Definition 5.1.1.1

- Especially, a kind of presheaf is the **representable presheaf**

$$F \cong h_X = Hom_C(-, X) : C^{op} \to Set \qquad U \mapsto Hom_C(U, X)$$

- The natural transformations induced by $F : X \to Y$ could be

$$h_F : h_X \to h_Y$$

And similarly

Definition 5.1.1.2 ...

5.1.2 Yoneda Embedding

Definition 5.1.2.1 Define:

$$Y : C \to Psh(C) \qquad Y \mapsto Hom_C(-, c)$$

which is called **Yoneda embedding**,

♠ **Proposition 5.1.2.2** Yoneda functor is:

- fully faithful;
- Preservation and Reflection the Isomorphsims;
- Dense(Density);
- Interaction With Density Comonads;
- Interaction With Codensity Monads;

∘ **Proof** ... □

Definition 5.1.2.3 The **Coyoneda embedding**

...

5.1.3 Yoneda's Lemma

▶ ***Theorem*** (Yoneda) Let $\mathcal{F} : C^{op} \to Set$ be a presheaf. We have following bijection

$$Nat(h_A, \mathcal{F}) \cong \mathcal{F}(A)$$

Natural in $A \in Obj(C)$. Especially,determining a natural isomorphism of functors

$$Nat(h__, \mathcal{F}) \cong \mathcal{F}$$

∘ **Proof** ... □

★ **Corollary 5.1.3.1** ...

∘ **Proof** ... □

★ **Corollary 5.1.3.2** (CoYoneda)

∘ **Proof** ... □

5.2 Restsricted Yoneda Embedding,Yoneda Extension

5.3 Representable Functors

5.3.1 Generalized Element

- `https://ncatlab.org/nlab/show/generalized+element`
- `https://www.voutsadakis.com/TEACH/LECTURES/CATEGORY/Chapter4.pdf?utm_source=chatgpt.com` P31

In category Set, the element can be seen as the maps

$$x : 1 \to A; A \in Ob(Set)$$

And in general

Definition 5.3.1.1 Let: C be an category and $A \in C$. Then, a morphism

$$x : S \to A$$

is a generalized element of A of shape S.

Especially, the generalized element is associated to the Yoneda embedding

$$Y : C \hookrightarrow [C^{op}, Set] \qquad X \mapsto GenEl(X)$$

Under this embedding, $GenEl(X) : C^{op} \to Set, Ob(C) \ni U \mapsto (U \to X)$.

5.3.2 Locally Representable Category

Definition 5.3.2.1 ...

5.3.3 Density Theorems

5.4 Isbell Duality

5.5 Closed Category

5.5.1 LaPlaza-Manzyuk Definition

Definition 5.5.1.1 A **closed category** is a category C with the following datas:

- A functor

$$[-,-] = \underline{Hom}(-,-) : C^{op} \times C \to C$$

  called the **internal hom-functor**;

- An object $I \in C$ called **unit object**;
- A natural isomrophism

$$i : Id_C \xrightarrow{\cong} [I, -]$$

- A transformation $j_X : I \to [X, X]$, the **extranatural** in X
- A transformation

$$L^X_{YZ} : [Y, Z] \to [[X, Y], [Y, Z]]$$

which is required satisfy the following axioms:

- The following diagram commutes for any X, Y:

$$\begin{array}{ccc} I & \xrightarrow{j_X} & [Y,Y] \\ & \searrow^{j_{[X,Y]}} & \downarrow L^X_{YY} \\ & & [[X,Y],[X,Y]] \end{array}$$

- $$\begin{array}{ccc} [X,Y] & \xrightarrow{L^X_{XY}} & [[X,X],[X,Y]] \\ & \searrow^{i_{[X,Y]}} & \downarrow [j_X, 1] \\ & & [I,[X,Y]] \end{array}$$

•
$$\begin{array}{ccc} [Y,Z] & \xrightarrow{L^I_{YZ}} & [[I,Y],[I,Z]] \\ & \searrow^{[1,i_Z]} & \downarrow{\scriptstyle [i_Y,1]} \\ & & [Y,[I,Z]] \end{array}$$

•
$$\begin{array}{ccc} [U,V] & \xrightarrow{L^Y_{UV}} & [[Y,U],[Y,V]] \\ \downarrow{\scriptstyle L^X_{UV}} & & \\ [[X,U],[X,V]] & & \downarrow{\scriptstyle [1,L^X_{YV}]} \\ \downarrow{\scriptstyle L^{[X,Y]}_{[X,U],[X,V]}} & & \\ [[[X,Y]],[X,U]],[[X,Y],[X,V]]] & \xrightarrow{[L^X_{YU},1]} & [[Y,U],[[X,Y],[X,V]]] \end{array}$$

- Finally,

$$\gamma : C(X,Y) \to C(I,[X,Y]) \qquad f \mapsto [1,f](j_X)$$

is a bijection.

Symmetric Closed Category

Definition 5.5.1.2 Furthermore, a **symmetric closed category** is additionally equipped with natural isomorphisms

$$C^{XY}_Z : [X,[Y,Z]] \cong [Y,[X,Z]]$$

Cartesian Closed Category

Remark Not 'Closed' Cartesian category!

Definition 5.5.1.3 The Cartesian category has:

- C has a terminal object 1;
- Finite/Binary Product $X \times Y$;
- There exists adjointness

$$- \times X \dashv [X,-]$$

i.e. $Hom(Z \times X, Y) \cong Hom(Z,[X,Y])$;
Such $[X,Y] \in C$ is called the **exponential object**.

Thus, we obtain those maps:

- $!_X : X \to 1$;
- $\Delta_X : X \to X \times X$;

And in Cartesian closed category:

- Weakening K:

$$K^X_Y : Y \to [X,Y]$$

- Contraction

$$T_Y^X : [X,[X,Y]] \to [X,Y]$$

♠ **Proposition 5.5.1.4** If C is a Cartesian closed category, then

$$X \times \sqcup_{i\in I} A_i \cong \sqcup_{i\in I} X \times A_i$$

This implies, that this is a distributive category.

∘ **Proof** We obtain, that

$$Hom(X \times \sqcup A_i, Y) \cong Hom(\sqcup A_i, [X,Y]) \cong \prod Hom_C(X \times A_i, Y) \cong Hom(\sqcup(X \times A_i), Y)$$

□

Some of examples:

♣ **Example 5.5.1.5**

- Category Set;

S-Combinator

Definition 5.5.1.6 Consider:

$$S_{Y,Z}^X : [X,[Y,Z]] \to [[X,Y],[X,Z]]$$

Operators C, S, T, K, have their correspondence in logic operation

- K: Weakening

$$K_Y^X : Y \to [X,Y] \Leftrightarrow K(y)(x) = y$$

- C: Exchange

$$C_{X,Y}^Z :\Leftrightarrow C(f)(y)(x) = f(x)(y)$$

- T contraction

...

- S:

5.5.2 Eilenberg-Kelly Definition

Definition 5.5.2.1 (Eilenberg-Kelly) Instead of γ, there exists an underlying-set functor

$$U : C \to Set$$

such that $U([X,Y]) = C(X,Y)$.Then isomorphism

$$C(X,X) = U([X,X]) \overset{Ui_{X,X}}{\to} U([I,[X,X]]) = C(I,[X,X]) \qquad 1_X \mapsto j_X$$

The wo are essentially equivalent.

5.5.3 Prounital Closed Category

Definition 5.5.3.1 Especially, when C is not required: A **prounitial closed category** is a category C together with the following data:

- A functor $[-,-] : C^{op} \times C \to C$,
- A functor $U : C \to Set$;
- A transformation $L^X_{YZ} : [Y,Z] \to [[X,Y],[X,Z]]$;
- A transformation $ev : U(X) \to C([X,Y],Y)$;

with following axiom:

-
-
-
-
-
-

5.5.4 Extra Natural Transformation

5.5.5 Reconstruction of Monoidal Structure

Suppose that $[A,[B,-]] : C \to C$ is representable as C-enriched functor, i.e.

$$F = [A,[B,-]] : C \to C$$

be a functor, then it is representable if there exists some $R \in C$ such that

$$[A,[B,X]] \cong [R,X]$$

Then, if we can find some R, then define it as $R = A \otimes B$, thus the followings are equivalnet:

- $[A,[B,-]]$ is enriched-representable;
- Exists $A \otimes B$, $[A,[B,X]] \cong [A \otimes B, X]$

Chapter 6 Limits

Set-Thoeretic Limits

Limit of Functors in an Arbitrary Category

Definition 6.0.0.1 (Limit of a functor, wth values in an arbitrary category) Now, take the notion of limit for functors

$$F^{op} : D^{op} \to \mathcal{C}$$

for $\mathcal{C}$ an arbitrary category, if we construct a certain presheaf $\mathcal{C}$, which we will call $\hat{lim}F$.
More precisely, using the **Yoneda embedding**

...

Then:

$$(\hat{lim})(c) := Hom_{Set^{D^{op}}}(pt, Hom_{\mathcal{C}}(c, F(-)))$$

Universal Cone

Definition 6.0.0.2 (Universal Cone) Let $F : D^{op} \to \mathcal{C}$ be a functor. Then, every object $c \in \mathcal{C}$, an element

$$* \to Hom(pt, Hom(c, F(-)))$$

is identiafied with a collection of morphisms $(c \to F(d))$.
For all $d \in D$, such that all triangles

$$c$$

$$F(d_i) \qquad\qquad F(d_j)$$

Such a collection of morphisms is called a **cone** over F.

6.0.1 Colimits

6.0.2 Compeleteness,Co-Compeleteness

Definition 6.0.2.1 ...

6.1 Product,Coproduct

6.2 Pullback and Pushout

6.3 Equalizer,Coequalizer

6.4 Special Limits

The reference is: Mac Lane.

6.4.1 Filtered Limits

6.4.2 Ends, Coends

Ends

Definition 6.4.2.1 ...

Coends

Ends with Parameters

Chapter 7 Localization of Categories

7.1 Multiplicative Sets

Definition 7.1.0.1 Let C be a category. Then a set of arrows S of C is called a **left multiplicative system** if it has following property:

-
-
-

Similarly, the right multiplicative system is well-defined.

7.1.1 Left Calculus of Fraction

7.2 Gabriel-Zisman Localization

Now we will introduce a starker localization,

Definition 7.2.0.1 (Gabriel-Zisman)Suppose $\mathcal{K}$ be an additive category, and $S \subseteq \mathcal{K}$ be a subclass:

- (FR1)S is closed under the composition of morphisms;
  $Id_X \in S, \forall X \in \mathcal{K}$;
- (FR2)We use $\Rightarrow$ to represent the morphisms in S:
  -
  -
- (FR3)

Especially, a Gabriel-Zisman system is called **text** if it has the property:

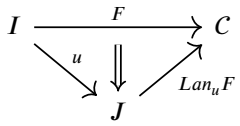
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Chapter 8 Kan Extension

8.1 Definition

Definition 8.1.0.1 Suppose: $u : I \to J, F : I \to \mathcal{C}$. Now

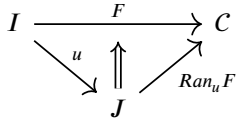
- The **left Kan extension** $Lan_u F$



  It satisfies the following property: For $G : J \to \mathcal{C}$,

$$Nat(Lan_u F, G) \cong Nat(F, G \circ u)$$

- The **right Kan extension** $Ran_u F$:



  with the following property: For $G : J \to C$,

$$Nat(Ran_u F, G) \cong Nat(G \circ u, F)$$

Thus,

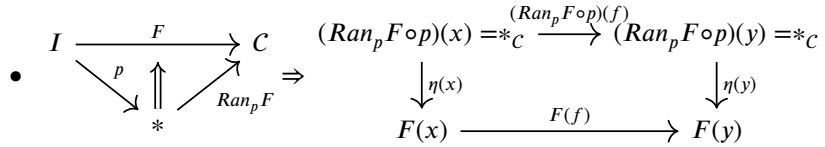
$$Lan_u : Hom(I, C) \to Hom(J, C)$$
$$Ran_u : Hom(I, C) \to Hom(J, C)$$

♠ **Proposition 8.1.0.2** Let $J = *$, then

- The left kan extension operation forms the colimit of a functor;
- The right Kan extension operation forms the limit of a functor.

∘ **Proof**

- •



  We obtain that

$$F(f) \circ \eta(x) = \eta(y) \circ (Ran_p F \circ p)(f)$$

  And such $Ran_p F \circ p(f) = id_{*_C}$, thus the diagram became the diagram of colimit;

- Similarly.

□

Presheaves

♠ **Proposition 8.1.0.3** ...

∘ **Proof** ... □

8.1.1 Pointwise Kan Extensions

▶ ***Theorem*** (Pointwise Kan Extension Formula) Suppose $u : I \to J$ be the functor between two small categories:

- If C is compelete, then the left Kan extension exists;
- If C is cocompelete, then the right Kan extension exists;

∘ **Proof**
Conversely similar. □

8.1.2 Kan Limits, and Other Categoricial Concepts

All concepts are Kan extensions ——Mac Lane

8.2 Derived Functors as Kan Extensions

Chapter 9 Monads, and Their Algebras

9.1 Monads

Definition 9.1.0.1 A **monad** on a category C consists of:

- An endomorphism $T : C \to C$;
- A **unit** is a natural transformation $\eta : id_C \Rightarrow T$;
- A **multiplication**: Natural transformation $\mu : T^2 \Rightarrow T$;

With the following diagrams commute in $Mor(C)$

$$\begin{array}{ccc} T^3 & \xRightarrow{T\mu} & T^2 \\ {\scriptstyle \mu T}\Downarrow & & \Downarrow{\scriptstyle \mu} \\ T^2 & \xRightarrow{\mu} & T \end{array} \qquad \begin{array}{ccccc} T & \xRightarrow{\eta T} & T^2 & \xLeftarrow{T\eta} & T \\ & {\scriptstyle id_T}\searrow & \Downarrow{\scriptstyle \mu} & \swarrow{\scriptstyle id_T} & \\ & & T & & \end{array}$$

i.e.

$$\mu \circ \mu T = \mu \circ T\mu$$
$$\mu \circ \eta T = \mu \circ T\eta = id_T$$

Commutative Monads

Definition 9.1.0.2 Especially, let (T, η, μ) be monad on symmetric monoidal category C, equipped with **left-length**

$$\sigma_{X,Y} : TX \otimes TY \to T(TX \otimes Y) \qquad X, Y \in Ob(C)$$

With braiding, we have **right-length** $\tau_{X,Y} = \sigma_{X,Y} \circ s(X, Y)$.
Then, T is **commutative** if the following diagram commutes for all objects X, Y of C

$$\begin{array}{ccc} & TX \otimes TY & \\ & & \\ T(TX \otimes Y) & & T(X \otimes TY) \\ & & \\ TT(X \otimes Y) & & TT(X \otimes Y) \\ & & \\ & T(X \otimes Y) & \end{array}$$

Definition 9.1.0.3 For such structure, take the notion

$$\nabla_{X,Y} : TX \otimes TY \to T(X \otimes Y)$$

which is called **monoidal multiplication**

9.2 Adjunction

9.3 Monadic Functors

9.4 Algebra over Monads

Definition 9.4.0.1 Let (T, η, μ)be a monad on category C. An **algebra over** T consists of

- Object A of C;
- Morphism $\alpha : T(A) \to A$ of C;
- (Unit Property)

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & T(A) \\ & \searrow^{id_A} & \downarrow{\scriptstyle \alpha} \\ & & A \end{array}$$

We will take the notion of C^T for the category of algebras over T.
Remark Such category is also called **Eilenberg-Moore category**, EM.

Homomorphism

Definition 9.4.0.2 Given two algewbras (A, α); (B, β), then a **T-algebra homomorphism** satisfies: The following diagram commutes

$$\begin{array}{ccc} TA & \xrightarrow{Tf} & TB \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

Free Algebras

Definition 9.4.0.3 Given a monad (T, μ, η), then for every object $X \in C$, TX has a T-algebra structure

$$\mu_X : T(T(X)) \to T(X)$$

T-algebra of this sort is called **free T-algebra**.

9.4.1 Eilenberg-Moore Category

Definition 9.4.1.1 Let (T, η, μ) be a monad, with $T : C \to C$ is an endofunctor with multiplication

$$\mu : TT \to T \qquad \eta : Id_C \to T$$

There is a forgetful functor

$$U^T : C^T \to C$$

Definition 9.4.1.2 (E-M Object)

♠ **Proposition 9.4.1.3** ...

∘ **Proof** ... □

Tensor Products

Definition 9.4.1.4 Especially, for the monad (T, μ, η) over a commutative monad $(C, \otimes, 1)$, and denote the multiplication of T by $\triangledown$. Then, the **tensor algebra** $A \otimes_T B$ is defined as coequalizer of diagram

$$T(TA \otimes TB) \qquad TT(A \otimes B) \qquad T(A \otimes B) \qquad A \otimes_T B$$

Bimorphisms, and Universal Property

Monoidal Closed Case

9.4.2 Bar Constructions

Definition 9.4.2.1 (Monoid Δ_a) Actually, Δ_a has following structure:

- Objects: $[n], n \in \mathbb{N}$;
- Morphisms: Order-Preserving Maps

and

- Tensor product: $[n] \otimes [m] := [m+n]$;
- Unit Object: $[0]$;
- Initial, Terminal Object: $[0], [1]$;

Actually, $(\Delta_a, +, [0])$ is a strict monoidal category.
Remark It is not strict: The category contains no isomorphism $[m]+[n] \xrightarrow{\simeq} [n]+[m]$.
And: $[1]$ is a monoid object. With

$$[n] = \sum [1]$$

So: Δ_a is free strict monoidal category, generated by a monoid object.

Now: Let E be a category with monad $(T, m : TT \to T, u : 1_E \to T)$, and category of T-algebras E^T, and a forget functor

$$U : E^T \to E$$

with $U \dashv F$. Such FU is a comonoid in $[E^T, E^T]$:

$$\delta : FU \Rightarrow FUFU \qquad \epsilon : FU \Rightarrow Id$$

With unit $\eta : Id_E \Rightarrow UF$, counit $FU \Rightarrow Id_{E^T}$, then $\delta = F\eta U : FU \to FUFU$.
Thus, we got a correspondence: Comonad on

Bar Constructions

Definition 9.4.2.2 There is a stroing monoidal functor

$$\Delta_a^{op} \xrightarrow{Bar_T} [E^T, E^T]$$

such that $1 \mapsto FU, [n] \mapsto [FU]^n$.

Actually: Let $G = FU$

$$Bar_T(A)_n := T^{n+1}(A)$$

And:

$$Bar_T(A) =$$

With evaluation for $A \in E^T$,

$$[E^T, E^T]$$

Then:

Definition 9.4.2.3

- A **bar construction** is

$$\Delta_a^{op} \to [E^T, E^T] \to E^T$$

- A **bar resolution** is the composite

$$\Delta_a^{op} \to E^T \to E$$

Now let $T = UF$,

...

Some special cases should be given here, to show that Bar construction works:

♣ **Example 9.4.2.4** (Classifying Space)
♣ **Example 9.4.2.5** (Group Cohomology)
♣ **Example 9.4.2.6** (Hochschild Homology)
♣ **Example 9.4.2.7** (Descending Theory of Module)
♣ **Example 9.4.2.8** (Adams Spectral Sequence)
♣ **Example 9.4.2.9** (Konzul Duality)

Chapter 10 Monoidal Category

10.1 Fundamentals

10.1.1 Monoidal Category

Definition 10.1.1.1 A **monoidal category** is a category $\mathcal{C} = (C)$ equpped with

- A functor
$$\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$$
out of the product cateegory of $\mathcal{C}$ with itself, called the **tensor product**.
- An object $1 \in \mathcal{C}$ called **unit object** or **tensor object**
- A natural isomorphism
$$\alpha : ((-) \otimes (-)) \otimes (-) \xrightarrow{\simeq} (-) \otimes ((-) \otimes (-))$$
called **associator**.
- A natural isomorphism: **Right Unitor**
$$\lambda : (1 \otimes (-)) \xrightarrow{\cong} (-)$$
- A natural isomorphism : **Left Unitor**
$$\rho : (-) \otimes 1 \xrightarrow{\cong} (-)$$

Such that

- (Triangle Identity) We have
$$(x \otimes 1) \otimes y \qquad\qquad x \otimes (1 \otimes y)$$
$$x \otimes y$$
- (Pentagon Identity)
$$(w \otimes x) \otimes (y \otimes z)$$
$$((w \otimes x) \otimes y) \otimes z \qquad\qquad w \otimes (x \otimes (y \otimes z))$$
$$(w \otimes (x \otimes y)) \otimes z \qquad\qquad w \otimes ((x \otimes y) \otimes z)$$

♣ **Example 10.1.1.2** (Monad) Especially, a **monad** is category $\mathcal{C} = (C, \otimes, 1, \lambda)$, with $(End(\mathcal{C}), id\mathcal{C}\circ, T, T)$

♣ **Example 10.1.1.3**

- Category $Vect_k$;

Strict Monoidal Cateogory

Definition 10.1.1.4 Instead of natural isomorphism, we have

...

10.1.2 Monoid

Definition 10.1.2.1 Given a monoidal category, then the **monoid** is the object $M \in C$, equipped with object $\mu : 1 \to M$ satisfying

$(M \otimes M) \otimes M$ $M \otimes (M \otimes M)$ $M \otimes M$

(Associative Law)

$M \otimes M$ M

$I \otimes M$ $M \otimes M$ $M \otimes I$

(Left and Right Unit Law)

M

10.1.3 Monoidal Functor

Definition 10.1.3.1 Let $(C, \otimes_C, 1_C)$ and $(D, \otimes_D, 1_D)$ be monoidal categories. A **lax monoidal functor** between them is a functor

...

together with coherence maps:

- $\eta : 1_D \to F(1_C)$;
- $\mu_{x,y} : F(x) \otimes_D F(y) \to F(x \otimes_C y)$, $x, y \in C$

Satisfies:

- (Associativity)
- (Unitality)

10.1.4 Dualizable Object

Definition 10.1.4.1 An object A in $(C, \otimes, 1)$ is **dualizable**, if it has an adjoint regarded as a morphism in the one-object delooping category BC corresponding to C.

Precisely, suppose $Ob(BC) = *$, and BC is a 2-category, C is a category, then

- $Hom(*, *) = C$;
- Given $A, B \in C$, then they should be seen as a 1-morphisms in BC:

$$A, B : * \to *$$

And morphism $f : A \to B$ became 2-morphism in BC.

- Especially, the composition in BC corresponds the tensor product in C:

$$\begin{aligned} & A \circ B \quad \text{in } BC \\ \Leftrightarrow\ & A \otimes B \quad \text{in } C \end{aligned}$$

And adjointness: For object $A \in C$ =1-morphism in BC there exists

$$coev: \quad ev$$

which satisfies **Zig-zag identities**

-
-

And:

Definition 10.1.4.2 (Terminologies)

-
-

10.1.5 Symmetric Monoidal Category

Symmetric Monoidal Category

Definition 10.1.5.1 A monoidal category is **symmetric** if it has a braiding.

Invertible Object

Definition 10.1.5.2 In a monoidal category, an **invertible object** is a dualizable object, such that unit and counit maps are isomorphsim(in BC)

Especially: For a symmetric monoidal category, we just need to verify the unit map.

Line Object

Definition 10.1.5.3 ...

Picard Groups

Definition 10.1.5.4 Given a (symmetric) monoidal category $(C, \otimes, 1)$, then the **Picard group** is the isomorphism classes of invertible objects, those that have an inverse under the tensor proudct.

Take the following notion

$$Pic(C) = \{[L] | L \text{ is inversable}\}$$

Remark Why is isomorphism class?

♠ **Proposition 10.1.5.5** C is symmetric$\Rightarrow$ $Pic(C)$ is abelian group.

∘ **Proof** ... □

Furthermore, we have **Picard groupoid**

Definition 10.1.5.6 A **Picard groupoid** is a **strict symmetric monoidal category** $(\mathcal{A}, \otimes, 1)$ in which:
Take the following notion

$$\mathbf{Pic}(C) := \text{Picard Groupoid}$$

The Picard groupoid is a kind of abelian 2-group. By definition we got

$$Pic(C) = \pi_0\mathbf{Pic}(C)$$

10.1.6 Closed Monoidal Category

Closed Monoidal Category

Definition 10.1.6.1 A **symmetric monoidal category** C is **closed**, if for all objects $b \in Ob(C)$, there exists adjunction

- Functor

...

- Right Adjoint Functor

...

Then, *hom*-isomorphism

$$Hom_C(X \otimes Y, Z) \cong Hom_C(X, [Y, Z])$$

10.1.7 Lax Monoidal Category,Lax Monoidal Functors

Lax Monoidal Category

Lax Monoidal Functors

Definition 10.1.7.1

Oplax Monoidal

10.1.8 Cartesian Monoidal Category

Definition 10.1.8.1 A **cartesian monoidal category** is a **monoidal category**, with the monoidal structure

$$(-) \otimes -(-) = (-) \times (-)$$

Cartesian Closed Category, ccc

Definition 10.1.8.2

- A **cartesian closed category,ccc** is a category with finite products which is closed with respect to its cartesian monoidal structure.
- The **internal hom** $[S, X]$ in a cartesian closed category is often called **exponentiation**, denoted by

$$X^S$$

the existence of $[-, -]$ is specified by the closed category.

- A **cartesian closed functor** between cartesian closed categories C, D is a functor $F : C \to D$ thath is product-preserving, and that is also expotential-preserving, i.e.

  ...

  corresponding to the composite

  ...

♠ **Proposition 10.1.8.3** Every ccc is self-enriched. (Definition in Next Section)

∘ **Proof** Given $(\mathcal{C}, \times, 1)$, and exponential objects

$$Y^X := [X, Y]$$

which is specified by

$$Hom_{\mathcal{C}}(A, Y^X) \cong Hom_{\mathcal{C}}(A \times X, Y)$$

$Hom(- \times X, Y)$ is representable. Then, define the enriched hom

$$\underline{Hom}_{\mathcal{C}}(X, Y) \in Y^X \in \mathcal{C}$$

- Composition

$$Z^Y \times Y^X \to Z^X$$

By adjunction $Hom_{\mathcal{C}}(Z^Y \times Y^X, Z^X) \cong Hom_{\mathcal{C}}((Z^Y \times Y^X) \times X, Z)$, the corresponding map could be given by

$$(Z^Y \times Y^X) \times X = Z^Y \times (Y^X \times X) \overset{id_{ZY} \times ev}{\to} Z^Y \times Y \overset{ev}{\to} Z$$

- Identity

$$1 \to X^X$$

We have $Hom(1, X^X) \cong Hom(1 \times X, X) \cong Hom(X, X)$. This is enriched identity.
which is specified

- Left, Right morphism trivial.

Thus, $C(X, Y)_{enriched} = Y^X$. □

★ **Corollary 10.1.8.4** $\mathbf{s}Set$ is self-enriched.

∘ **Proof** Given by the strucuture

$$(\mathbf{s}Set, \times, \Delta[0])$$

□

10.2 Braiding

Definition 10.2.0.1 Given a monoidal category $(\mathcal{C}, 1, \otimes, \alpha, \lambda, \rho)$, the **braid structure** is the isomorphism between binary functors

$$c(X,Y) : X \otimes Y \xrightarrow{\sim} Y \otimes X; X, Y \in Ob(\mathcal{C})$$

Such that the following diagram commutes:

...

Definition 10.2.0.2 (Symmetric Monoidal Category) If

$$c(X,Y) \circ c(Y,X) = id_{X \otimes Y} \qquad \forall X, Y \in Ob(\mathcal{C})$$

Then, the category is called **symmetric monoidal category**.

♠ **Proposition 10.2.0.3** (Yang-Baxter Equation) Suppose $\mathcal{C}$ be a monoidal category with braid, then the following diagram commutes:

...

∘ **Proof** ... □

Especially:

★ **Corollary 10.2.0.4** If $\mathcal{V}$ is strict monoidal category with braid, then the diagram is equivalent to say:

$$(c(Y,Z) \otimes id_X)(id_Y \otimes c(X,Z))(c(X,Y) \otimes id_Z) = (id_Z \otimes c(X,Y))(c(X,Z) \otimes id_Y)(id_X \otimes c(Y,Z))$$

∘ **Proof** ... □

10.2.1 Braid Category

We will introduce Braid category, the contents of braid group, we will study it in an another topic.

10.3 Enrichment of Monoidal Category

Instead of Hom-Set, we want to observe Hom-Objects.

Definition 10.3.0.1 Let V be a monoidal category, then a **small V-category** or **V-enriched category** is:

- A **set** $Obj(C) \in Set$, the set of objects;
- For each pair

$$(a, b) \in Ob(C) \times Ob(C)$$

There exists an object $C(a,b) \in Obj(V)$ called **hom-object**

Sometimes, the collection is called **enriched Hom-object** and can be written as

$$\underline{Hom}_C(X,Y)$$

- For each order triple (a, b, c), $[\circ_{a,b,c} : C(b, c) \otimes C(a, b) \to C(a, c)] \in Mor(V)$, called **composition morphism**
- Identity Element: For $a \in Obj(\mathcal{C})$, ther e exists a morphism $[j_a : 1 \to C(a, a)] \in Mor(V)$ called **identity element**
- The following diagrams commutes:

$$\begin{array}{ccc} (C(c,d) \otimes C(b,c) \otimes C(a,b)) & & C(c,d) \otimes (C(b,c) \otimes C(a,b)) \\ \\ C(b,d) \otimes C(a,b) & & C(a,d) \otimes C(a,c) \\ \\ & C(a,d) & \\ C(b,b) \otimes C(a,b) & & C(a,b) \otimes C(a,a) \\ \\ & C(a,b) & \\ \\ I \otimes C(a,b) & & C(a,b) \otimes I \end{array}$$

10.3.1 Enriched Functor

Definition 10.3.1.1 Let C, D be enriched in monoidal category V, and **enriched functor** $F : C \to D$ consists of:

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10.3.2 Enrichment by Lax Monoidal Functor

Definition 10.3.2.1 Let V be a monoidal category, then V-category can be viewed, as a set $X \in Set$ together with a **lax monoidal functor** $\Phi = \Phi_d$, of the form

$$\Phi : V^{op} \to Set^V \xrightarrow{d} Set^{X \times X}$$

With:

- d is induced by the pairing $(a, b) \mapsto C(a, b)$;
- $Set^{X \times X}$ has the convolution structure

$$(F \odot G)(x, z) := \sqcup_{y \in X} F(y, z) \times G(x, y)$$

- $\Phi(v)_{x,y} := hom_V(v, d(x, y))$;
  Under the composition,

$$Hom(v, C(x, y)) \times Hom(v', C(y, z)) \to Hom(v \otimes v', C(x, y) \otimes C(y, z)) \xrightarrow{(Hom(1,comp))} Hom(v \otimes v', C(x, z))$$

We have

$$\Phi_d(v) \odot \Phi_d(w) \to \Phi_d(v \otimes w)$$

which is a lax monoidal structure.

10.4 MacLane's Coherence Theorem

According to the definition of monoidal category, the associator, left unitor and right unitor are given by the equivalence relation. But furthermore, a question is: We have

$$((X \otimes Y) \otimes Z) \otimes W \sim (X \otimes (Y \otimes Z)) \otimes W \sim X \otimes ((Y \otimes Z) \otimes W) \sim X \otimes (Y \otimes (Z \otimes W))$$
$$((X \otimes Y) \otimes Z) \otimes W \sim (X \otimes Y) \otimes (Z \otimes W) \sim X \otimes (Y \otimes (Z \otimes W))$$

And the isomorphisms are may be not trivial. If it is strict, then the process is unncessary.

▶ ***Theorem*** (MacLane)

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∘ **Proof** ... □

★ **Corollary 10.4.0.1** (Stritification Theorem)For every monoidal category, there exists a canonical isomorphism

$$\mathcal{C} \simeq \mathcal{C}_{str}, \alpha, \lambda, \rho = 1$$

Chapter 11 Enriched Category

Reference:

- https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/kelly-book.pdf

Chapter 12 Bicategory,2-Category

12.1 Bicategory

Definition 12.1.0.1 A **bicategory** B consists of:

- A collection of objects called 0-cells;
- For each pair 0-cells (x, y), a category

$$B(x, y)$$

whose objects are called **morphisms** or **1-cells** and whose morphisms are called **2-morphisms** or **2-cells**

- For each 0-cell x, a distinguished 1-cell

$$1_x \in B(x, x)$$

called the **identity morphism** or **identity** 1**-cell** at x.

- For 0-cell x, y, z, horizontal composition

$$\circ : B(y, z) \times B(x, y) \to B(x, z)$$

called **horizontal composition**

- For each pair of 0-cell , natural isomorphism in $B(x, y)$ called **unitors**

$$\begin{pmatrix} f \mapsto f \circ 1_x \\ \theta \mapsto \theta \circ 1_x \end{pmatrix} \cong id_{B(x,y)} \cong \begin{pmatrix} f \mapsto 1_y \circ f \\ \theta \mapsto 1_{1_y} \circ \theta \end{pmatrix} : B(x, y) \to B(x, y)$$

As tradition, denoted by $\lambda_{x,y}, \rho_{x,y}$.

- Associators:Consider the composition funcotr

$$c_{x,y,z} : B(y, z) \times B(x, y) \to B(x, z) \qquad (g, f) \mapsto g \circ f$$

Then, there exists an natural isomorphism between functors:

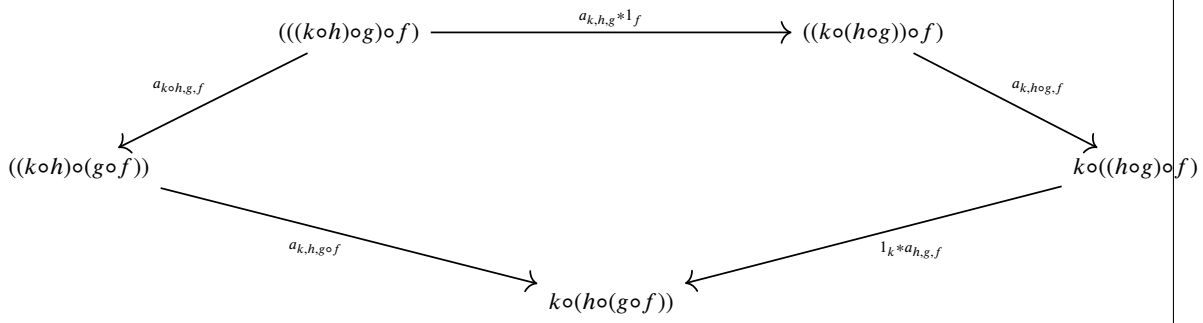
$$a_{w,x,y,z} : c_{w,x,z} \circ (c_{x,y,z} \times 1_{B(w,x)}) \Rightarrow c_{w,y,z} \circ (1_{B(y,z)} \times c_{w,x,y})$$

They are functors $B(x, y) \times B(y, z) \times B(z, w) \to B(x, w)$. And componentwise, they are 1-morphisms:

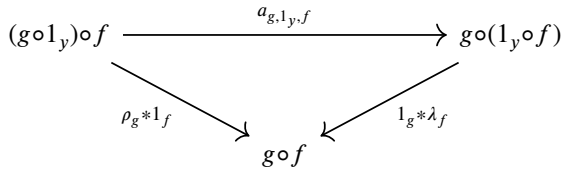
$$a_{f,g,h} : (h \circ g) \circ f \overset{\cong}{\Rightarrow} h \circ (g \circ h)$$

Such that:

- The pentagon identity is satified by the associators;



- The Triangle Identity is satisfied by unitors.



Actually, it is a weak 2-category.

12.1.1 2-Functors

Definition 12.1.1.1 ...

Pseudo-Natural Transformations

Definition 12.1.1.2 ...

Biadjunction

Definition 12.1.1.3 Given 2-categories A, C, with two functors

$$F : A \to C \qquad U : C \to A$$

Then, a **biadjunction** is given by: For each object $a \in A; c \in C$ there exists equivalence of categories

$$C(Fa, c) \cong A(a, Uc)$$

which is pseudo-natural both a and c.

12.2 2-Category